Research Article

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Technological Training in Artificial Intelligence and Human Capital Skills Development: Impact on Labor Productivity in the Textile Industry

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Abstract

This research aimed to analyze the relationship between technological training in artificial intelligence and the development of human capital skills and their impact on labor productivity in the textile industry, from the employees' perspective. A quantitative approach was used, with a non-experimental, simple correlational, and cross-sectional design, applying a structured survey to 216 workers. The results revealed a positive and significant correlation between the two variables studied. Three dimensions of the first variable were considered: deep learning, robotics, and virtual assistants, with deep learning being the most influential factor. Furthermore, it was identified that technological training plays a mediating role in the effective adoption of advanced technologies, contributing to the development of digital skills and the improvement of job performance. The study concludes that greater employee training in artificial intelligence drives a significant increase in labor productivity. Therefore, it is recommended to strengthen continuous training programs and promote the strategic use of technology to enhance innovation and productivity.

Keywords: *Automation, Training, Artificial Intelligence, Textile Industry, Productivity.*

1. Introduction

In recent decades, global society has undergone a radical transformation driven by the advancement of digital technologies. Among these, artificial intelligence (AI) has emerged as a disruptive force with great potential to redefine production, organizational, and educational

processes (Xu et al., 2021). In particular, technological training and the development of human capital skills can be a critical mediator influencing the effectiveness with which AI is adopted and leveraged in work contexts (Nurlia, Daud & Rosadi, 2023). The link between continuous technological training and the workforce can transform the processes of technical knowledge transfer within organizations, reshaping how new employees are trained and developed (Spitzer, Kühl, & Goutier, 2022). International organizations project that the adoption of AI can contribute to a reversal of productivity growth, although they emphasize that the benefits depend on the ability to integrate complementary assets, including digital infrastructure and the ongoing training of human capital (OECD, 2024). Consequently, training and skills development emerge as necessary conditions to convert technological adoption into sustained performance improvements (Zhang & Lee, 2025).

This research is technically justified by the need to understand and measure how training in AI-based technologies and skills development influences levels of labor productivity, particularly in industrial sectors that require high operational efficiency. The integration of AI into business processes has positive effects in areas such as time management, task automation, predictive analytics, and strategic decision-making. According to a McKinsey & Company report (2023), organizations that have adopted AI solutions in their value chains have reported productivity increases of up to 20%, primarily due to the intelligent automation of repetitive functions and the use of algorithms to optimize workflows.

Companies seek to identify their employees' skills and strengths in order to align them with their daily processes and activities, thereby facilitating business management. The intention is to foster the ongoing training of employees in the use of AI tools and their development, so that these tools serve as a support for increased productivity rather than being perceived as an obstacle within the organization's strategic processes.

The study by Noy and Zhang (2023) demonstrates, through a randomized, pre-registered experiment, that access to generative AI coupled with ongoing training substantially increases productivity in professional writing tasks. With a sample of 444 participants, those who received a crash course and practiced with the tool completed tasks in approximately 40% less time and produced higher-quality output (18% higher than the control group, which was evaluated blindly). The effect was most pronounced among workers with lower initial performance, suggesting that AI, facilitated by minimal training, reduces performance gaps by automating stages such as the first draft and freeing up time for revision.

Liu, Zhang, and Wei (2025) conducted a study on generative AI literacy (GAIL), which shows that it improves job performance when strengthened with ongoing training. This training covers five areas: technical foundations, advanced prompting, critical evaluation of outputs, job application, and responsible and ethical use. The study validates this framework with psychometric models and structural analyses, demonstrating that GAIL significantly increases creative self-efficacy. This self-efficacy acts as a mediating mechanism that transforms training into real performance improvements. Furthermore, simply providing access to the tool is insufficient; deliberate micro-training sessions are required. Short, structured programs that teach how to request, evaluate, and adapt results using AI produce measurable improvements in work quality and speed—in other words, in productivity.

Yunus, NM, & Zouya, L. (2024) conducted research analyzing the impact of continuous training in artificial intelligence and skills development on labor productivity in the manufacturing and service sectors in China, using registered application patents as a reference. The results showed a more significant effect on the productivity of the manufacturing sector compared to the service sector. Furthermore, a higher proportion of highly skilled workers was observed in manufacturing. Similarly, Atoche (2023) carried out a study highlighting the positive impact of AI training on increasing business productivity, noting substantial improvements in both sales levels and customer satisfaction. Likewise, Barrios, HJ, and Mio Trinidad, JA (2023) conducted research

focused on the strategic use of AI training for optimizing business performance, the results of which showed significant sales growth, thus confirming the effectiveness of this technology as a business growth tool.

The study conducted by Nurlia, Daud, and Rosadi (2023) explored how artificial intelligence training influences employee productivity. The results highlight that this positive impact does not occur automatically but depends primarily on two factors: the training and skills development that employees receive to use these tools, and the organization's ability to adapt to technological changes. When adequate AI training is provided and organizational adjustments are promoted, productivity improves significantly. Conversely, the lack of these elements can limit the benefits of AI or generate resistance among staff.

The study by Al Naqbi, Bahroun and Ahmed (2024) reviews recent research on how training in generative artificial intelligence (such as ChatGPT and other similar tools) and skills development can improve the productivity of human capital at work. Through an analysis of multiple studies, the authors conclude that these technologies allow for the automation of repetitive tasks, improve decision-making, and support creative processes, where their positive impact depends on the level of technological integration in organizations and the constant training of employees.

Bravo and Iturralde (2024) conducted an analysis of how robust training in artificial intelligence is being implemented in the management of textile companies. Their study reveals that this software technology allows for the automation of key processes, improves decision-making, and optimizes resource planning. It also contributes to more efficient inventory management and traceability throughout the production chain. The authors emphasize that the impact of artificial intelligence is not uniform, as it depends on employees' level of digitalization and knowledge of advanced technologies. Organizations with an innovative mindset and a culture open to change reap greater benefits. In contrast, those with traditional structures face more barriers to its adoption.

Ruiz Toabanda (2025) conducted a study analyzing the impact of ongoing AI training on the labor productivity of companies in Guayaquil, identifying opportunities and challenges in its implementation. Three main objectives were established: to diagnose the technological and cultural barriers to AI adoption, to examine its effect on operational efficiency and productivity, and to determine successful integration strategies. The findings reveal that training and developing employees' AI skills improves productivity and efficiency in processes such as customer service, inventory management, and data analysis. However, companies that have successfully implemented AI have developed strategies for training their staff, investing in technology, and implementing a phased integration. The study concludes that, to maximize the benefits of AI, it is essential to overcome cultural and technological barriers, promoting the ongoing development of human talent, fostering an organizational culture oriented toward innovation, encouraging continuous learning, and ensuring the appropriate infrastructure for its effective adoption.

Canossa-Montes de Oca and Peraza-Villarreal (2024) A study was conducted analyzing the role of artificial intelligence (AI) in human talent management (HTM) within organizations. Key findings include, firstly, identifying challenges to integrating AI into the company, such as promoting adaptive leadership, establishing a technological infrastructure, digitally transforming and adapting roles, preparing human talent in AI, and building a technological culture. Secondly, the study analyzes potential opportunities within HTM functions, from strategic human talent planning to developing human competencies through AI-driven initiatives. Emphasis is placed on the human requirements for effective adaptation, such as digital skills and intelligence, creativity and socio-emotional skills, promoting lifelong learning, and investing in human capital. The research presents an optimistic view of how employees can leverage AI to optimize and enhance the competitiveness of organizations.

2. Methodology

This research is applied in nature, as it seeks to address specific problems in real-world contexts, employing scientific knowledge as a basis for improving processes, making informed decisions, and proposing practical solutions (Hernández-Sampieri & Mendoza, 2021). Regarding the methodological approach, the quantitative paradigm was chosen, which focuses on the collection and analysis of numerical data to answer precise research questions and test previously established hypotheses. This paradigm aims to ensure the objectivity, replicability, and generalizability of the findings through the use of standardized tools and statistical methods (Hernández-Sampieri & Mendoza, 2021). The methodological design is non-experimental and of a simple correlational type, which allowed for the examination of the possible association between the variables considered in the study without intentionally manipulating any of them. Using inferential statistics, underlying population information was obtained for this relationship, under a cross-sectional design that involved the three dimensions of variable 1 with variable 2 of the study. This design is characterized by not including the intentional manipulation of the independent variables by the researcher, and is frequently used in descriptive and correlational cases, as it allows observing, describing and analyzing phenomena as they occur in their natural context (Hernández-Sampieri & Mendoza, 2021).

Regarding the study variables, the first, training and development of skills in artificial intelligence (AI), has transformed various industrial sectors, including textile manufacturing, this change being motivated by the need to improve operational efficiency, reduce associated expenses and raise the quality standards in the products, achieving greater competitiveness in the global market (Chui et al., 2018), which is evidenced in the way in which this technology is applied in the textile sector, covering from the design stages to the logistics processes. Its incorporation has enabled the introduction of innovative solutions to the challenges of the sector, especially through the use of generative algorithms and neural networks, which facilitate the creation of personalized patterns and the prediction of market trends. Artificial intelligence, a branch of technology that, through computer networks, allows for the efficient handling and analysis of large volumes of data, becomes more accessible and easier to manage when integrated into neural networks. These networks simulate the capabilities of human intelligence, including deep learning, robotics, and virtual assistants. The second variable is productivity, which is essential in both the economy and business management, as it evaluates how efficiently resources are converted into goods and services that, when properly managed, benefit the company by improving productivity (Ramírez Méndez et al., 2022). Regarding this other variable, productivity encompasses not only the efficiency of processes but also factors such as innovation, strategic human capital management, and the ability to adapt to change. A key approach to increasing productivity is the efficient use of technological tools and automated processes, and the implementation of digital systems have proven effective in reducing costs and improving operational efficiency from the dimensions of efficiency, effectiveness, and Lean Six Sigma (LSS) tools, which are methods, techniques, and resources used to improve processes, eliminate waste, and reduce variability in an organization's operations.

The target population for this study consisted of 490 employees from various departments. Inclusion criteria required employees with at least six months of service with the organization. Conversely, exclusion criteria included employees with less than six months of service. To determine the sample size, a special statistical formula for small groups was used, based on a 95% confidence level, an expected response rate of 50%, and allowing for a 5% margin of error. Under these parameters, the calculated sample size was 216 employees. The population refers to the entire group of individuals who share common characteristics relevant to the study, while the sample represents a portion of that population that is analyzed to draw inferences (Otzen & Manterola, 2017).

Regarding the technique used, it is the survey, which involves the development of four main components: sampling methods, questionnaire design, application of reliable interviews, and descriptive and inferential statistical analysis (Zúñiga & Cedeño, 2023). The data collection instrument is a questionnaire composed of 30 items organized on a Likert scale, with ordinal categories ranging from the lowest to the highest level. According to Hernández-Sampieri and Mendoza (2021), the Likert scale is a data collection instrument that allows for measuring the attitudes, perceptions, or opinions of respondents through a set of declarative items, in which the participant is asked to express their level of agreement or disagreement on an ordinal scale.

The analysis of the data collected in this research was carried out at two levels: descriptive and inferential, using SPSS statistical software version 26.0, in accordance with the quantitative approach and correlational design of the study (Hernández-Sampieri & Mendoza, 2021). First, descriptive statistics were applied to characterize employees' perceptions of technological training and skills development in artificial intelligence regarding human capital and productivity. For this purpose, absolute and relative frequencies (percentages) were calculated for each dimension of the variables: Deep Learning, Robotics, and Virtual Assistant in the case of AI; and Efficiency, Effectiveness, and LSS Tools in the case of Productivity. Subsequently, inferential statistics were applied to determine the relationship between the study variables. Since the data were collected using Likert scales, considered ordinal in nature, a non-parametric technique was chosen. Kendall's Tau-b test was used, which is recommended for determining the correlation between ordinal or non-normally distributed variables, as it assesses the degree of agreement or association between two variables and allows for hypothesis testing (Field, 2024). Furthermore, the precision of the instruments was determined using the Internal Consistency test, which was applied to 216 employees. The results exceeded the critical level of 70%. The artificial intelligence skills training and development instrument obtained a Cronbach's Alpha of 0.907 (90.7%), while the Productivity instrument reached 0.835 (83.5%). Both values significantly exceed the critical level of 0.70 proposed by Cronbach (1951), demonstrating acceptable reliability.

3. Results

The results of the study are presented through a descriptive and correlational analysis of AI and productivity. The statistical software SPSS v26.0 was used, and Kendall's Tau-b test was applied to determine the relationship between the two variables.

Table 1. Frequency by levels of perception on technological training in artificial intelligence and development of human capital skills.

Level	Frequency	(%)
High	**120**	**56.0**
Half	**80**	**37.0**
Low	**16**	**7.0**
Total	**216**	**100.0**

Source: Own elaboration.

Of the total number of employees (216), 56% showed a high perception of technological training in artificial intelligence and skills development, reflecting a positive assessment of its implementation. 37% showed a medium perception, and only 7% a low perception, indicating that the majority recognize technological training and skills development in AI as a useful and beneficial tool in their work environment.

Table 2. Technological training in artificial intelligence and development of human capital skills and productivity according to employee perception.

		Productivity			
		High	Half	Low	Total
Technological training and skills development in artificial intelligence	High	40	5	0	45
	Half	10	74	9	93
	Low	0	18	60	78
Total					216

Source: Own elaboration

Of the 216 employees surveyed, 88.9% of those with a high perception of technological training in artificial intelligence and skills development exhibited high productivity, while 11.1% showed average productivity. Among those with an average perception, 79.6% demonstrated average productivity, 10.8% high productivity, and 9.7% low productivity. In the group with a low perception, 76.9% showed low productivity and 23.1% average productivity. In summary, the results confirm that greater training and perception in AI are associated with higher levels of workplace productivity.

Table 3. Correlation between technological training in artificial intelligence and the development of human capital skills and productivity

		Productivity
Technological training in artificial intelligence and human capital skills development	Correlation coefficient (tb)	0.637
	Significance (p-value)	0.000
	N	216

Source: Own elaboration.

A correlation coefficient of $\tau = 0.637$ was identified, reflecting a significant positive correlation between the variables examined. The significance level achieved ($p < .001$) confirms that this association is statistically significant.

Table 4. Correlation between the Deep Learning dimension and Productivity

		Productivity
Deep learning	Correlation coefficient (tb)	0.511
	Significance (p-value)	0.000
	N	216

Source: Own elaboration

A correlation coefficient of $\tau = 0.511$ was obtained, demonstrating a positive association between the analyzed variables. Furthermore, the significance level $p < .001$ confirms the statistical validity of this relationship.

Table 5. Correlation between the Robotics dimension and Productivity

		Productivity
Robotics	Correlation coefficient (tb)	0.326
	Significance (p-value)	0.000
	N	216

Source: Own elaboration.

The correlation coefficient obtained was τ = 0.326, demonstrating a moderately positive association between the analyzed variables. Furthermore, the significance level p < .001 supports its statistical relevance.

Table 6. Correlation between the Virtual Assistant dimension and Productivity.

		Productivity
Virtual Assistant	Correlation coefficient (tb)	0.506
	Significance (p-value)	0.000
	N	216

Source: Own elaboration.

A correlation coefficient of τ = 0.506 was obtained, reflecting a strong positive relationship between the variables studied. Furthermore, the statistical significance (p < .001) confirms that this relationship is statistically supported.

4. Discussion

The findings of this study align with extensive empirical evidence demonstrating that technological training in artificial intelligence (AI) and the development of human capital skills are decisive factors in improving labor productivity and organizational efficiency. In the context of digital transformation, companies seek to identify their employees' competencies and align them with production processes. This requires training programs that enable employees to understand, apply, and leverage AI tools not as a threat, but as a strategic support for business management. Furthermore, we observed a positive and significant correlation between employees' technological training and the development of artificial intelligence skills and productivity (τ-b = 0.637; p = 0.000), confirming that greater AI training increases efficiency, goal achievement, and process optimization in the textile sector. This finding coincides with previous studies that highlight the role of AI as a driver of automation and competitiveness (Yunus & Zouya, 2024; Atoche, 2023). Regarding the dimensions, deep learning showed a moderate correlation (τ-b = 0.511), reinforcing its role as a driver of innovation and predictive analytics (LeCun, Bengio & Hinton, 2015); robotics showed a positive but weaker relationship (τ-b = 0.326), explained by structural limitations and technological investment; while virtual assistants showed a significant correlation (τ-b = 0.506), highlighting their impact on administrative efficiency and organizational communication (Pereira et al., 2023).
These results confirm that technological training in AI and skills development boost productivity through knowledge that enables automation, error reduction, and improved decision-making.

However, when interpreting these findings, some inherent limitations of the study must be considered. Its cross-sectional design prevents the establishment of causal relationships, and the

sample, restricted to the textile sector in East Lima, limits generalizability to other contexts. Future studies should incorporate longitudinal and comparative analyses between sectors or regions, as well as include mediating variables such as digital leadership or organizational culture (Liu, Zhang & Wei, 2025). Furthermore, it is recommended to employ causal models and mixed-methods approaches to delve deeper into how technological training in artificial intelligence transforms labor productivity (Chen, 2025; Nurlia, Daud & Rosadi, 2023). These considerations will strengthen the empirical evidence and generate a more comprehensive understanding of the impact of AI on business efficiency and competitiveness.

5. Conclusions

The study concludes that AI technology training and skills development for a well-prepared workforce have a clearly positive impact on labor productivity, especially when their implementation is strategically aligned with employees' daily operational functions and coherently integrated with organizational objectives. This technology enables the automation of repetitive tasks, reducing the operational workload and freeing up time for higher-value activities such as innovation and continuous improvement.

Its use also facilitates process standardization, early detection of inefficiencies, and real-time monitoring of work performance by providing access to accurate and up-to-date data. Furthermore, ongoing AI training enables data-driven decision-making and fosters a culture of continuous improvement. Furthermore, the results of the statistical analysis showed a significant association between its dimensions (deep learning, robotics, and virtual assistants) and levels of labor productivity. Deep learning was identified as the dimension with the greatest impact due to its ability to process large volumes of unstructured data, learn autonomously, and continuously improve its performance through feedback. This allows for the identification of hidden patterns, the prediction of behaviors, and the optimization of real-time decision-making. Its implementation facilitates the intelligent automation of complex processes, reducing errors, increasing operational accuracy, and generating a competitive advantage by enabling more agile responses, product customization, and rapid adaptation to changes in demand or market conditions.

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